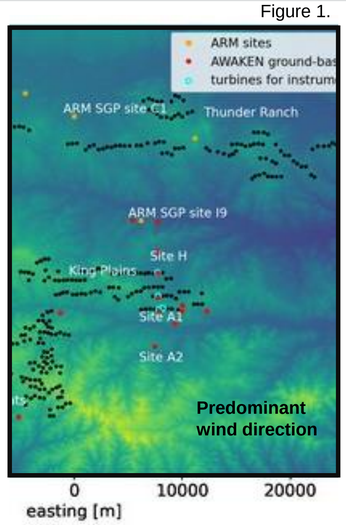
AWAKEN Campaign Observations of Precipitation from Hub Height, Inflow, and Outflow Regions of the King Plains Wind Farm

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Motivation

Impacts with hydrometeors contribute to turbine blade leading edge erosion. We investigate how wake effects may affect droplet characteristics using disdrometer measurements at AWAKEN.

Figure 1.



Observational Set Up

- In conditions of southerly winds, FOV = 160 – 200º (Fig. 1)
  - Site A1 = Inflow, ground
  - Site H = Outflow, ground
  - Turbine H8 has disdrometer installed at hub height
- Instrumentation
  - Parsivel Laser disdrometer
  - Climavue50 compact weather station (Fig. 2)
  - Turbine SCADA data

Figure 2.



Comparison to Literature

Disdrometers are sophisticated sensors that measure the numbers, diameters, and fall speed of droplets (Fig. 3), compared to simpler instruments that only record rain rate. Several empirical expressions have been proposed for estimating the distribution of sizes among a population of droplets for given rain rate. Common expressions imperfectly capture the size distributions we observe (Fig. 4), which have a relative abundance of droplets around 1 mm. Downwind sites also tend to have a greater than expected prevalence of large drops several mm in size. In future work, we will assess the physical processes driving these differences in droplet size distributions.

Figure 3.

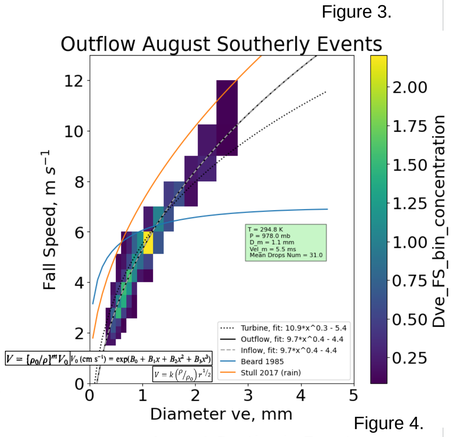
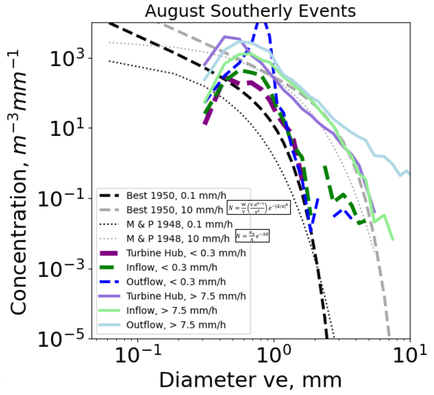
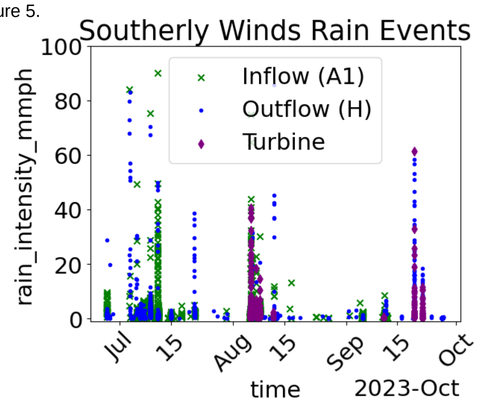


Figure 4.



Observations June – September 2023

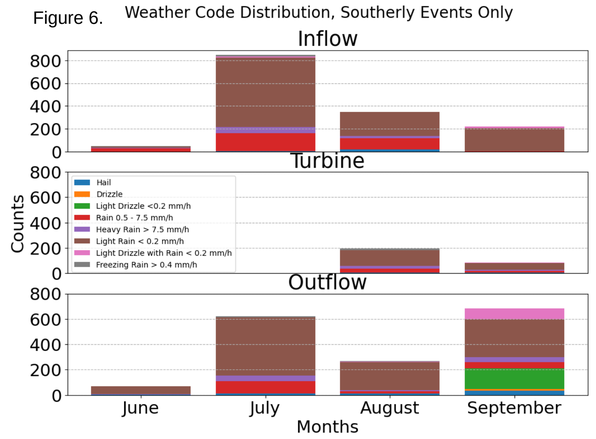
Figure 5.



Precipitation events have a high spatial and temporal variability, making comparisons among just a few measurement sites challenging. Simultaneous precipitation at all three sites is relatively infrequent (Fig. 5). The probability of precipitation at site H (outflow) given rain at site A1 (inflow) within 1 hour is 75%. It is 75% at H given T (turbine) and 71% at T given A1 (August only).

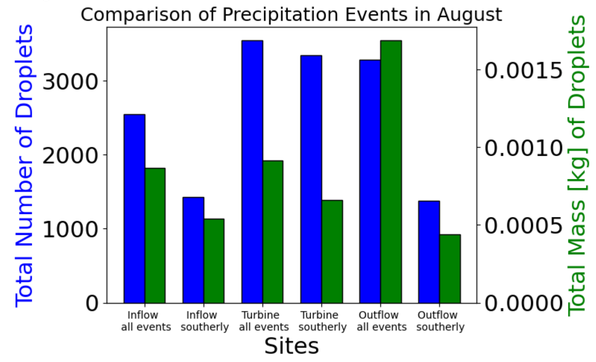
A larger ratio of total number to total mass indicates a larger abundance of smaller droplets. We see the largest ratio of these smaller droplets at the turbine site (Fig. 7). The instrument was installed on the turbine nacelle just behind the spinning blades (Fig. 8)

Figure 6.

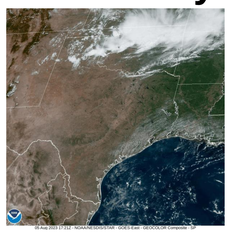


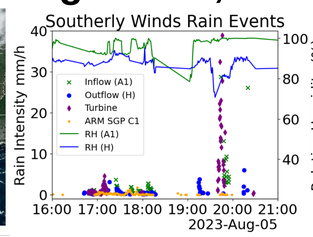
During the study period, hail, rain, and drizzle were recorded at each site (Fig. 6). The inflow site detected more cases of rain while the other two sites detected more light rain. The outflow site detected more cases of solid precipitation in September.

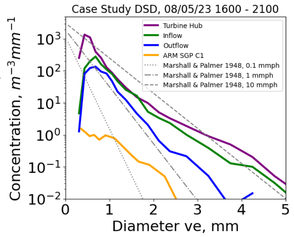
Figure 7.



Case Study August 5th, 2023





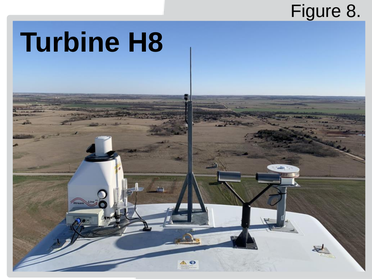


Southerly frontal system with blanket stratus. We observed occasions when precipitation was recorded at hub height but not at the surface. Less precipitation recorded at outflow site.

Conclusion

For the first time, a laser disdrometer was installed at hub height during the AWAKEN campaign at the King Plains wind farm. The peak in small diameters during low rain intensity events at the outflow site may be associated with wake-generated turbulence of the wind farm. While more work is needed to ascertain the robustness of these results, changes to the droplet distribution downstream must be understand to accurately predict leading edge blade erosion for downwind turbines.

Figure 8.





Campaign Webpage

Data Hub

For additional information, contact:
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